

Frequency distribution tables

By size and month

Task 1: Calculate the confidence intervals for men shoes sales in the USA, this time based on a bigger sample - 2015-2016.

Task 2: Compare the results with the confidence intervals when the sample was based just on 2016.

Solution:

United States, 2015													United States, 2016												Mean	Standard error
US	1	2	3	4	5	6	7	8	9	10	11	12	1	2	3	4	5	6	7	8	9	10	11	12	15-'16	15-'16
6	0	0	0	0	3	1	1	3	5	4	0	0	4	1	3	1	3	3	3	4	3	7	3	0	2.17	0.39
6.5	4	1	0	1	0	0	2	3	0	1	3	3	3	2	0	1	0	0	1	7	2	1	2	1	1.58	0.34
7	0	0	0	0	1	2	1	0	1	2	2	3	0	0	1	0	6	4	4	2	3	0	0	0	1.33	0.34
7.5	0	1	2	1	0	0	3	2	3	2	2	2	3	2	3	1	7	0	7	3	4	6	1	1	2.33	0.41
8	5	3	1	0	6	6	4	0	5	6	3	3	7	9	7	3	12	2	9	4	7	5	2	6	4.79	0.60
8.5	4	5	3	1	2	11	6	6	4	12	4	2	12	12	8	8	15	9	17	17	6	9	10	6	7.88	0.94
9	10	14	7	9	17	16	20	21	13	17	10	12	17	13	13	11	21	22	25	30	26	25	13	10	16.33	1.26
9.5	18	16	23	15	16	16	19	26	25	17	35	24	19	25	27	24	26	33	25	47	31	44	37	26	25.58	1.77
10	8	13	13	15	7	20	22	18	22	15	11	14	17	26	26	19	16	31	25	24	23	31	15	20	18.79	1.33
10.5	7	13	8	8	15	10	16	8	16	16	17	14	13	16	22	14	28	19	18	15	19	21	16	10	14.96	1.02
11	5	5	4	7	5	5	4	6	3	5	5	10	5	16	13	10	10	11	15	8	9	7	6	7	7.54	0.72
11.5	3	3	2	3	6	9	6	6	3	8	3	7	4	3	6	3	3	5	6	4	5	12	13	5	5.33	0.58
12	1	0	0	3	1	4	1	2	4	3	6	3	3	0	0	4	4	4	3	12	4	9	2	1	3.08	0.58
13	0	0	1	0	1	3	1	0	0	3	0	1	1	1	2	0	3	2	1	0	0	4	3	2	1.21	0.26
14	1	2	1	2	1	0	3	3	0	1	2	2	2	6	3	3	5	3	2	1	0	1	2	1	1.96	0.30
15	0	0	0	0	1	0	0	1	1	0	1	1	0	0	0	1	1	0	4	0	0	0	0	2	0.54	0.19
16	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.00	0.00
Total	66	76	65	65	82	103	109	105	105	112	104	101	110	132	134	103	160	148	165	178	142	182	125	98		

Task 1

Margin of error	95% CI		Number of pairs, math round
15-'16	15-'16		
0.81	1.35	2.98	3
0.70	0.88	2.29	2
0.70	0.63	2.03	2
0.85	1.48	3.18	3
1.24	3.55	6.03	6
1.95	5.92	9.83	10
2.61	13.72	18.94	19
3.65	21.93	29.24	29
2.74	16.05	21.53	22
2.11	12.85	17.07	17
1.49	6.05	9.03	9
1.21	4.13	6.54	7
1.20	1.88	4.28	4
0.53	0.68	1.74	2
0.62	1.34	2.57	3
0.39	0.15	0.94	1
0.00	0.00	0.00	0

	15-'16
n	24
t _{95%,23}	2.07

Task 2

Number of pairs, round up	Number of pairs (lesson)	Difference Lesson - Exercise
3	4	1
3	3	1
3	3	1
4	5	2
7	8	2
10	13	3
19	23	4
30	36	7
22	26	4
18	21	4
10	12	3
7	8	1
5	6	2
2	2	0
3	4	1
1	1	0
0	0	0

The numbers are close, however most shoe sizes seem overestimated when we base the prediction only on one year. This may be because 2016 was a very good year for sales, while 2015 wasn't.